Physical Consequence, Not Localization, Governs Terrain-Induced Target Re-observability

First Author, Second Author, Third Author
Department, Organization, City, Country

first.author@example.edu

A robot that has already observed a target can lose sight of it because its own motion and the intervening terrain block the line of sight. Restoring visibility is not a search problem: the robot knows approximately where the target is, and that knowledge is still not enough. This paper asks what information and what action reasoning terrain-induced target re-observability actually requires, and answers by elimination. Paired worlds that share terrain, pose, heading and a pixel-identical current observation still require opposite re-observation actions, with disjoint successful action sets in eight of eight pairs, so no memoryless rule can solve both; a second family additionally shares the target memory and still requires different optimal actions. Over 240 (state, action) pairs, 43.1 percent of the actions a geometric visibility oracle endorses fail when executed, while the same oracle placed at the pose the robot actually reaches agrees with the physical outcome 97.9 percent of the time, so the missing quantity is the physical consequence of an action rather than its visibility. On 2240 physically executed (state, action) pairs, adding the candidate action to observation plus memory raises re-observation AUC from 0.854 to 0.934, whereas adding the current observation to the memory buys 0.002; a planner built on the resulting action-conditioned consequence model raises RSR@10 from 0.250 to 0.792 and RSR@1 from 0.167 to 0.708 without privileged information. Three negative results bound how such a model may be used. Anchor recovery under perfect localization still fails physically in six of twenty-four terrains. A safety consequence head that is accurate and made better calibrated exactly where the decision threshold acts yields zero planning gain. Lookahead of two, three and five steps is not significantly better than one step on any budget while costing 1.8 to 2.1 times the motion. All results are in simulation with a vendored third-party locomotion policy; we claim the formulation, the benchmark, the failure analysis and the identification of the missing quantity, not a world model.

Keywords—target re-observability, terrain-induced occlusion, action-conditioned consequence prediction, partial observability, legged locomotion, benchmark, failure analysis

I. Introduction

Robots operating in natural terrain rarely enjoy a clear view of what they care about. A legged platform descending i nto a crater, a rover traversing a ridge line, or a manipulator r eaching behind a boulder all face the same structural problem: the terrain determines what can be seen from where, and the r obot's own motion is what changes it. Partial observability here is not sensor noise but a deterministic consequence of geometry, and therefore predictable in principle - but only if t he robot reasons about how its next action will change what it can see.

This paper studies a specific instance of that problem. The robot has already observed a target. Later the target is no longer visible, not because it moved or the sensor failed, but because the robot's own position and the intervening terrain n ow block the line of sight. The robot must choose an action t hat makes the target visible again. We call this target re-observability, and it differs from classical search in one decisive way: the robot has information about where the target is, and that information is still not enough.

The natural intuition that moving toward a target makes it more likely to be seen fails under terrain. Measured along a traverse toward a target beyond a ridge, the visibility sequence is not a single transition from visible to hidden but alternates: stairs_up V X V X, trench V X V X, pit V X V X, barrier V X V X, waves X V X V X, perlin X V X V X, basin X V X, perlin_rough X V X V X, where V marks a position from w hich the target is visible and X one from which it is blocked. The number of transitions ranges from two to four on a single traverse. A policy that assumes monotonicity is not merely i mprecise; it is aimed at the wrong objective.

Prior work supplies necessary components but addresses d ifferent questions. Lost-target search maintains a belief over w here the target went and selects search actions to reduce it [1], [2]. Occlusion-aware active perception and next-best-view s elect viewpoints by predicting visibility or information gain f rom a candidate pose, typically using geometry rather than execution [3], [4]. Object permanence filters maintain a belief through periods of non-observation [5]. Embodied visual tracking and target-loss recovery is an established direction with its own benchmarks and systems [6], [7], and work in that line already combines action-conditioned world models with model-predictive control for recovering a lost target [8], [9]. None of this is unstudied, and we do not claim otherwise. What is missing is a unified analysis: these lines are evaluated in their own settings, with their own observation and action i nterfaces and their own budgets, so the boundaries between t hem are not visible. We place the same decision rules on one task - terrain fixed, target static, action set identical, only the d ecision rule differing - and ask where and why each breaks.

This paper makes the following contributions. First, a formulation of terrain-induced target re-observability as an action-selection problem, with the non-monotonicity of visibility along a traverse as its defining property. Second, a procedurally generated benchmark of eight terrain-induced o cclusion mechanisms, organised by physical mechanism and r eal-world analogue rather than as a difficulty ladder, with paired worlds that are provably identical in the current observation. Third, a frozen observation and action interface u nder which every decision rule is evaluated on the same terrain budget and termination set, with all comparisons reported as paired bootstrap intervals. Fourth, a measured identification o f the missing quantity, together with three negative results showing that accurate prediction is not the same thing as decision usefulness.

II. Related Work

Lost-target search maintains a belief over target location a nd plans search actions to reduce uncertainty [1], [2]. It answers which search action best reduces uncertainty, but not what to do when terrain makes the correct re-observation action point away from the belief. In our paired worlds the belief is correct and the action that restores visibility lies on t he opposite side.

Occlusion-aware active perception and next-best-view score candidate viewpoints by predicted visibility or information gain [3], [4]. Candidate poses are typically evaluated kinematically, which assumes that an action takes t he robot exactly where its nominal displacement says. We m easure that this assumption fails, and that the visibility model itself is not the problem.

Object permanence maintains a belief through non-observation and acts on the filtered estimate [5]. It answers w here the target probably is; it does not establish that knowing where the target is determines what to do. Our second pair family isolates exactly that gap.

Embodied visual tracking and target-loss recovery is an e stablished direction [6], [7], with work already combining action-conditioned world models and model-predictive control for target recovery [8], and recent work coupling visibility with reachability and feasibility [9]. Our focus is narrower and physical: which quantity must be predicted for a known target to be made visible again under terrain-induced o cclusion, and how that prediction should enter a decision.

III. Problem Formulation

Let O_t be the observation at time t, comprising pose, proprioception and a short action history; M_t the target memory, that is where and when the target was last seen; and a_{t:t+H} a sequence of physical actions. Write V_{t+h} in {0,1} for target visibility after executing that sequence. The t ask is

$$a^*_{t:t+H} = \operatorname{argmax} P(V_{t+H} = 1 \mid O_t, M_t, a_{t:t+H}) \quad (1)$$

The target is lost when V_t = 0 after having been observed, and re-observed when there exists h <= H with V_{t+h} = 1. Physical safety is deliberately not part of this definition: Section VI shows that safety is a dimension of the consequence model rather than part of the task statement, and mixing it into (1) would confound the two.

The benchmark reports a difficulty curve rather than a single success flag, because the quantity of interest is how m any physical actions are needed to regain sight. For episode i and budget K,

$$r_i(K) = 1[\text{exists } h <= K : V_{i,t+h} = 1], \quad RSR@K = (1/N) \sum_i r_i(K) \quad (2)$$

Note what the definition does not require: visibility to be h eld. Visibility under these mechanisms is non-monotonic, and the task is to regain sight within a budget, not to keep it. The implementation is audited against (2) rather than assumed to match it: the metric is evaluated directly on the recorded per-step visibility trace, and a contract test asserts that this agrees with reading the termination state, so the headline number cannot drift if the episode semantics change.

Two companion metrics are reported and no single scalar combines them. The re-observation time T_reobs is the elapsed time from loss to the first re-observation. The action c ost C_action is cumulative base-path length, averaged over e very episode rather than only the successful ones so that exhausting the budget does not look cheap.

IV. Method

A. Visibility state and the missing quantity

H1, H2 and H3 below remove the simple explanations one at a time: the current observation does not determine the action a correct target memory does not determine it either, and geometric visibility does not predict physical re-observation. The quantity that remains is the action-conditioned physical c onsequence. Rather than learning visibility end to end, we exploit the third result: since the visibility oracle is accurate a t whatever pose the robot actually reaches, the model predicts the executed body-frame pose delta and the already-validated oracle is applied to the predicted pose. This keeps the validated part validated and localises the remaining error.

B. Candidate actions and the consequence model

Candidate actions are ten fixed locomotion primitives, each of the same simulated duration so that one step is one u nit of budget. The consequence model is trained on 2240 physically executed (state, action) pairs, that is 24 terrains by 10 states by 10 actions, with positive rate 0.143. Features are the pose, the projected gravity vector, a short pose history, the target memory expressed in body coordinates with an explicit presence flag, and the action. Targets are the executed pose d elta and whether the target was re-observed. Splits are by terrain and never random, so a terrain family cannot appear o n both sides.

C. Closed-loop action selection

The planner scores every candidate action with the model, takes the argmax of the predicted re-observation probability, a nd replans after each executed step, giving a receding-horizon controller over the learned model. An episode terminates at t he first re-observation, at a fall, or on leaving the terrain.

V. Experimental Setup

A. Benchmark

The suite comprises eight terrain-induced occlusion mechanisms, listed in Table I and organised by physical mechanism rather than by surface name. Each is a situation in which a robot that has already seen a target loses sight of it a nd must regain sight by moving. The mechanisms span five r oughly orthogonal axes: occlusion direction, elevation dependence, spatial scale, visibility transition frequency, and t errain spatial complexity. A mechanism is a point on an axis rather than a member of a category: pit and basin are both spatial scale, at opposite ends of it.

Fig. 1. Terrain-Induced Occlusion Mechanisms

Mechanism	Axis	I
barrier	occlusion direction	wa
trench	elevation	dit
stairs_up	elevation	s
pit	spatial scale	
basin	spatial scale	
waves	visibility dynamics	d
perlin	terrain complexity	na
perlin_rough	terrain complexity	

Two framing rules are enforced mechanically rather than l eft to intent. The taxonomy is checked for the wording challenging, hard, difficult or adversarial terrain and refuses t o render if it appears, because the difficulty is in the target's v isibility and not in the driving. And perlin and perlin_rough are not claimed to be real surfaces: Perlin noise is used as a p rocedural representation of spatially correlated natural elevation variation. All terrain is built as explicit box slabs rather than as a height field, so that the analytic visibility oracle and the collision surface are the same object.

B. Baselines and metrics

Every baseline implements the core decision mechanism o f a line of work rather than a full reproduction. Reactive always drives forward. HeadingLock holds the heading recorded at the loss moment. HistoryOnly aims at the remembered target. LostTargetSearch faces the belief and closes on it. ObjectPermanence runs a constant-velocity belief filter. NBV re-projects the believed target from a kinematic p ose and is given the exact height field, since an active-perception method is allowed a map. Oracle physically rolls o ut every candidate and is a privileged upper bound, never a d eployable method.

Metrics are RSR@K as defined in (2), T_reobs, and C_action. Every proportion carries a Wilson score interval rather than a normal approximation, because several results sit at or near zero or one. Every comparison is paired: since all p olicies run on the same terrains, a difference in RSR is tested by resampling terrains with replacement and examining the d istribution of the paired difference.

VI. Results

A. The current observation does not determine the action

We construct paired worlds that share terrain, robot pose, heading and a pixel-identical current observation, but in which the target was last seen from opposite sides. Across eight pairs the rendered camera images are identical with mean and maximum absolute pixel difference both zero, the optimal re-observation action is opposite in the two worlds in eight of eight, and the successful action sets are disjoint in eight of eight. Because any function of the current observation takes t he same value in both worlds, no memoryless policy can solve both. On sixteen paired-history worlds the five classical rules score zero of sixteen.

B. A correct target memory is also not sufficient

A natural reply is to remember where the target is. We therefore construct a second family of pairs sharing the current observation and the reconstructed target memory, differing o nly in terrain hidden inside the shadow of a visible occluder, which is invisible from the decision point and was never traversed. Over six such pairs the rendered observations are p ixel-identical in six of six, the target memory is identical in six of six, and the optimal action differs in four of six. Two properties bound the claim. The successful action sets overlap, so this does not show that no fixed rule can succeed; it shows that a correct memory does not determine the optimal action. And on these pairs all six classical policies choose the same a ction and fail in both worlds, so the construction is an information-level result and not a behavioural one.

C. Geometric visibility is not physical re-observability

For every candidate action at every decision state we compute two quantities. G(s,a) asks whether the target would be visible if the robot reached the ideal pose the action nominally produces, using a kinematic displacement and the l ine-of-sight oracle. P(s,a) asks whether the target is actually r e-observed when the primitive is executed. Over 240 pairs, the confusion matrix is [G=0,P=0] 171, [G=0,P=1] 11, [G=1,P=0] 25, [G=1,P=1] 33. That is, 43.1 percent of the actions geometry endorses fail physically, and the disagreement is not one-sided: geometry also under-predicts in 4.6 percent of cases. Placing the same oracle at the pose the robot actually r eaches makes it agree with the physical outcome in 97.9 percent of pairs. The pose prediction error is larger on the failing cell than on the agreeing cells by a paired bootstrap difference of 0.28 m with a 95 percent interval of 0.11 to 0.45 m, decomposed into 0.52 m forward, 0.34 m lateral, 0.40 m in height and 5.5 degrees of yaw. No active-perception method i s mis-implemented here; candidate viewpoints scored kinematically are simply a different quantity from where the robot ends up.

D. Information ablation

Table II ablates the predictor input with terrain held out. A dding the candidate action raises AUC by 0.080, while adding the current observation to the memory raises it by 0.002. The information is in the action, not in seeing more. On the sixteen paired-world decision states the model reaches 0.8125 accuracy and switches its chosen action in eight of eight pairs which is what a memoryless rule provably cannot do.

Fig. 2. Information Ablation, Leave-One-Terrain-Out, 2240 Samples

Inputs	ROC-AUC	PR-AUC
O	0.759	0.360
M	0.852	0.639
O + M	0.854	0.600
O + M + a	0.934	0.748

E. Analytic versus learned consequence

The learned pose model is compared against the analytic n ominal displacement, which is what a kinematic predictor assumes. Mean absolute error falls from 0.469 to 0.148, a factor of 3.2, with per-component ratios of 5.5 for forward, 2. 9 for lateral and 3.7 for yaw. The analytic model is nevertheless better for height, at 0.090 against 0.149, because it predicts no change and changes are small. The correct statement is that the learned model wins on the components t hat decide success, not that learning is better in all dimensions.

F. Consequence prediction converts into task performance

Table III reports the multi-step protocol over 24 terrains with identical budget and termination set for every method. R SR@10 rises from 0.250 to 0.792 and RSR@1 from 0.167 to 0.708 with no privileged information. By contrast, no classical baseline significantly outperforms any other: relative to HistoryOnly, Reactive, HeadingLock, LostTargetSearch and ObjectPermanence each differ by 0.000 with interval [0, 0], t hat is, they succeed and fail on exactly the same terrain set. N BV differs by 0.042 with interval [0.00, 0.125] and p = 0.356, so it cannot be claimed to be better. Only the privileged references separate.

Fig. 3. Multi-Step Episode Protocol, 24 Terrains

Method	RSR@1	RSR@3
six classical baselines	0.167	0.250
Learned consequence	0.708	0.708
Oracle (privileged)	0.833	0.875
Oracle search (privileged)	0.750	1.000

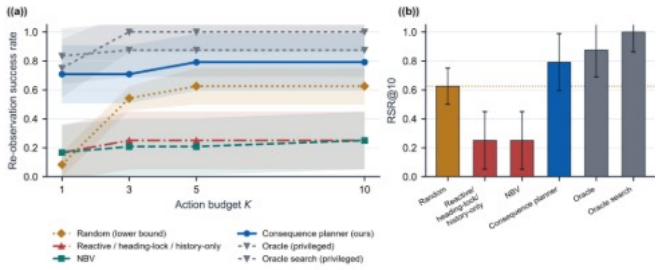


Fig. 4. Fig. 1. Re-observation performance and the baseline ladder, over 24 terrains with a 10-step budget. (A) Re-observation success rate against action budget. Intervals are Wilson 95 percent intervals, except for Random, which is a mean over three seeds and whose band is the spread across those seeds. (B) Success at the full budget with Wilson 95 percent intervals. Random, which never commits to a single action, sits above every classical rule; the consequence planner is higher still, at lower motion cost. The privileged oracles are reference bounds and are not deployable methods.

G. *Cross-mechanism generalisation*

The unit of generalisation here is the mechanism, not the t errain. An earlier predictor held out a single terrain, which for the sixteen paired-history worlds still leaves fifteen members of the same mechanism in training; that measures within-mechanism generalisation and cannot answer this question. W e therefore hold out an entire mechanism family at once. The learned pose model beats the analytic baseline on all five held-out mechanisms, at a mean error of 0.329 against 0.491, so the causal structure action to future physical state does transfer. A ccuracy nevertheless degrades relative to the within-mechanism reference, where mean pose error is 0.196 and m ean AUC 0.897 against 0.810 across mechanisms. The occlusion-direction fold reaches an AUC of 0.533, but that fold also removes 1680 of 2240 training rows, so it confounds the mechanism shift with a large reduction in data and cannot be attributed to mechanism transfer alone.

VII. Failure Analysis and Discussion

A. *The task is not localisation recovery*

The most damaging reading of this benchmark is that it reduces to state recovery: save the pose at which the target w as last seen and use odometry to return to it. We implement t hree such baselines - steering back to the last-visible pose, dead-reckoned backtracking along the recorded walk-in, and r eturning to that pose before advancing along the original sight ray - under perfect, moderate and realistic odometry drift. Perfect odometry does not solve the task: the best anchor-recovery baseline reaches RSR@10 of 0.250, the same as the classical rules and far below the consequence planner. More p ointedly, six of its twenty-four failures under perfect localisation are physical terminations, meaning the robot falls or leaves the terrain while returning to a pose it knows exactly. Perfect and moderate drift produce identical results and realistic drift changes RSR@10 by 0.000, so the failure is not an odometry-accuracy problem but the physical-consequence gap reached from an independent direction. Knowing exactly where the robot was does not imply that re-observation is physically executable.

B. *Failure type changes with planner capability*

Table IV classifies failures. Classical policies predominantly fail because they select ineffective actions, whereas the learned consequence planner substantially reduces action-selection failures and instead exposes physical termination failures. This shift is itself evidence that the modelled quantity is the right one and that the remaining gap concerns the coverage of the consequence model rather than candidate diversity. Our first version of this taxonomy recorded falls and boundary exits as action-selection failures, which is an attribution error: the action was chosen on a defensible prediction and what failed was the prediction of stability.

Fig. 5. Failure Classification Under Protocol B, 24 Terrains

Planner	Action (B)
HeadingLock / HistoryOnly / LostTargetSearch / ObjectPermanence	18
Reactive	17
NBV	16
Learned consequence	1
Oracle	3

C. *Better prediction is not better decisions*

Two negative results bound how a consequence model m ay be used, and both concern the prediction-to-decision step r ather than prediction quality. Adding a safety consequence head raises the privileged Oracle with ground truth from 0. 3884 to 0.4107, so the term is worth having, and the learned head is accurate with ROC-AUC 0.900. Yet the planner gets worse at every penalty weight. Fixing the obvious explanation does not help: the head's overall calibration error is 0.042 while its error in the band where a threshold actually acts is 0. 197, a factor of 4.7, and temperature scaling halves that to 0. 099 and produces exactly zero planning improvement. Across raw, temperature-scaled and isotonic variants no operating point at any threshold or coverage beats not using safety at all.

A dependence analysis over the same 2240 samples explains why. Visibility and safety are negatively coupled, w ith an odds ratio of 0.35 and a 95 percent interval of 0.249 to 0.492, so the actions that reveal the target are less safe than those that do not, and the coupling persists within six of the t en individual actions, so it is not merely induced by action choice. But it almost never creates a decision conflict: safe actions are abundant at 9.17 per state and visible actions scarce at 1.43, and only five of 224 states have a visible action and a safe action but none that is both. The headroom for any joint decision is therefore bounded by that 2.2 percent conflict set, which is smaller than the noise in any learned head, and the j oint head, a product of the two marginals, and a conformal filter all fail to improve on the visibility-only planner.

Longer lookahead fails in the same shape. Intermediate states in a chained rollout come from the learned model and n ever from ground-truth stepping, and only the first action of e ach plan is executed. Against a horizon of one step, paired success differences are minus 0.208 with interval minus 0.458 to 0.042, 0.000 with minus 0.208 to 0.208, and minus 0.125 with minus 0.333 to 0.083 for horizons of two, three and five. No horizon is significantly better on any budget, and every h orizon costs 1.78 to 2.09 times the motion. Lookahead does r escue two terrains that genuinely require multiple steps, which confirms that re-observation is intrinsically sequential, but it b reaks eight terrains that one step solves. Model-error accumulation is not the explanation: the measured one-step p ose error is lowest at a horizon of two and that horizon performs worst. The mechanism is that a chained score taking a maximum over steps is monotonically optimistic in the horizon, so a longer lookahead offers more opportunities to find a sequence the model confidently believes will glimpse t he target, and those are precisely the confidently wrong

predictions. Longer rollout is not automatically beneficial when consequence uncertainty accumulates.

D. Limitations

All results are from simulation with a vendored third-party locomotion policy on a single controller, and no sim-to-real c laim is made. The oracle is a privileged upper bound and the search reference is incomplete: an earlier version of this work judged four terrains unsolvable and beam search subsequently solved all four with two- to three-step sequences, so mistaking a planner's defect for a property of the benchmark nearly put a spurious ceiling into the results. The learned consequence model is evaluated with terrain held out, which is generalisation within a mechanism family. The target memory is a structured location representation reconstructed without n oise, so belief-estimation failure is excluded by construction. The action set is ten fixed primitives and whether discretisation affects the paired-world conclusions has not been tested. A binary fall label proved unusable, with zero falls in 720 single-action rollouts, so stability is represented by boundary exit and a continuous margin, which means the learned safety head does not cover the event it is named after. Finally, the paired-world construction is available only in mechanisms that can express a directional aperture; five mechanisms where this is structurally impossible are documented in the benchmark definition.

VIII. Conclusion

We asked what information and what action reasoning a r obot needs in order to regain sight of a target that terrain has occluded. The current observation is not sufficient, because w orlds with pixel-identical observations can require opposite a ctions with disjoint successful sets. A correct target memory i s not sufficient either, because worlds identical in observation and in memory can still require different actions. Geometric visibility is not sufficient, because 43.1 percent of the actions it endorses fail physically while the same oracle at the executed pose agrees with physics 97.9 percent of the time. W hat is missing is the action-conditioned physical consequence: adding the candidate action to observation plus memory raises predictive AUC by 0.080 while adding the observation to the memory raises it by 0.002, and a planner built on the resulting model raises RSR@10 from 0.250 to 0.792 without privileged information.

Three negative results bound how such a model may be u sed, and all concern the prediction-to-decision step rather than prediction quality. Perfect localisation does not guarantee physical re-observability. A safety head that is accurate and m ade better calibrated at the decision boundary produces zero planning gain. Deeper lookahead is not significantly better on any budget, costs nearly twice the motion, and rescues two g enuinely sequential terrains while breaking eight that one step solves. The compact statement of the paper is therefore that o bservation plus memory does not imply the optimal action, a nd the reason is the step in between: action to future physical state to future visibility must be modelled explicitly, must be action-conditioned, and must then be turned into a decision r ather than a threshold.

IX. References

[1] S. Radmard and E. A. Croft, “Active target search for high dimensional robotic systems,” Autonomous Robots, 2016

[2] S. Radmard, D. Meger, E. Law, and E. A. Croft, “Resolving occlusion in active visual target search of high-dimensional robotic systems,” IEEE Trans. Robotics, 2018.

[3] OA-NBV authors, “OA-NBV: Occlusion-aware next-best-view planning for human-centered active perception on m obile robots,” arXiv:2603.11072, 2026.

[4] LEAP-NBV authors, “LEAP-NBV: Lightweight edge active-perception for foundation-model next-best-view p lanning,” arXiv:2609.23974, 2026.

[5] Y. Peng et al., “Object permanence filter for robust tracking with interactive robots,” in Proc. IEEE ICRA, 2 024.

[6] S. Wang et al., “TrackVLA: Embodied visual tracking in the wild,” in Proc. Machine Learning Research, vol. 305, 2025.

[7] EVT-Bench authors, “EVT-Bench: A benchmark for embodied visual tracking,” arXiv:2506.01600, 2025.

[8] CST-WM authors, “CST-WM: A causally structured world model for embodied visual tracking,” arXiv:2609. 06302, 2026.

[9] Visible-Reachable authors, “Visible-reachable workspace for perception-aware humanoid design,” arXiv:2609. 08905, 2026.